

Supplementary Materials

Table S1: Summary of connected component sizes

Statistic	Value
Max size	13,873
Min size	2
Mean size	5.50
Median size	2.0
Component Size Distribution	
Component Size	Number of Components
13,873	1
53	1
35	1
34	1
33	1
30	1
27	1
24	1
23	2
22	1
21	1
20	4
19	1
18	4
17	6
16	4
15	8
14	7
13	12
12	12
11	17
10	33
9	27
8	48
7	75
6	117
5	212
4	383
3	1115
2	3383

Table S2: Summary of Community Detection Results

Cluster	Nodes / Edges	Average Node Degree	Top Three PIs (Normalized Degrees)
1	847 / 1295	3.06	RAMACHANDRAN, VASAN S (6.30) SESHADRI, SUDHA (6.30) MESCHIA, JAMES F (5.04)
2	841 / 1131	2.69	SCAMPAVIA, LOUIS DANIEL (8.80) NDHLOVU, LISHOMWA C (7.27) HORVATH, TAMAS L (5.24)
3	779 / 1123	2.88	MADABHUSHI, ANANT (7.68) OGEDEGBE, OLUGBENGA G. (5.30) MASON, CHRISTOPHER EDWARD (4.82)
4	778 / 1041	2.68	GRAY, NATHANAEL SCHIANDER (6.32) WILSON, DAVID M (5.30) LIN, HENING (4.78)
5	776 / 1141	2.94	XU, XIANGMIN (6.88) YEO, EUGENE WEI-MING (5.97) REN, BING (5.51)
6	769 / 1115	2.90	ZHANG, RONG (5.86) BINDER, ELLEN F (5.86) IRIMIA, DANIEL (4.41)
7	766 / 1228	3.21	BOOKHEIMER, SUSAN Y (5.93) BARCH, DEANNA (5.93) VAN ESSEN, DAVID C (5.12)
8	762 / 1089	2.86	CHUNG, WENDY K (7.21) KIZER, JORGE R (6.19) HOROWITZ, CAROL R (4.15)
9	743 / 1210	3.26	POWERS, ALVIN C (8.78) KIM, SEUNG K (5.02) KOENEN, KARESTAN C (5.02)
10	739 / 1196	3.24	PERICAK-VANCE, MARGARET A. (7.75) HAROUTUNIAN, VAHRAM (5.61) ROUSSOS, PANAGIOTIS (5.61)
11	733 / 1148	3.14	KENNY, EIMEAR ELIZABETH (6.52) EICHLER, EVAN (5.28) CLARK, ANDREW G (4.87)
12	723 / 1046	2.89	DEBAUN, MICHAEL R. (7.19) WEISS, MITCHELL J (4.81) SNYDER, MICHAEL P. (4.33)
13	723 / 1091	3.02	LI, DEBIAO (7.05) ZHANG, JIANYI (4.70) LENBURG, MARC ELLIOTT (4.70)
14	716 / 1097	3.07	WOODRUFF, PRESCOTT G (5.68) WENZEL, SALLY E (5.68) LEVY, BRUCE D (4.86)
15	715 / 1207	3.38	PLEVRITIS, SYLVIA KATINA (6.06) DE KONING, HARRY J (6.06) MANDELBLATT, JEANNE (5.34)
16	713 / 1169	3.28	ELLISMAN, MARK H (5.62) DICKERSON, BRADFORD C. (5.24) LIANGPUNSAKUL, SUTHAT (5.24)
17	711 / 1092	3.07	FIGUEIREDO, JANE C. (6.76) LE MARCHAND, LOIC (5.16) HAIMAN, CHRISTOPHER ALAN (4.77)

Cluster	Nodes / Edges	Average Node Degree	Top Three PIs (Normalized Degrees)
18	705 / 989	2.81	LEWIS, JASON S. (6.40) SHAW, DANIEL S (5.35) TERRY, MARY BETH (4.83)
19	334 / 448	2.68	DIAMOND, MICHAEL S (8.77) LI, JUN (4.21) FREMONT, DAVED H. (3.20)

Table S3: Top 10 representative PIs in each cluster based on normalized degree

Cluster	Top Ten PIs (Normalized Degrees)
0	RAMACHANDRAN, VASAN S (6.30) SESHADRI, SUDHA (6.30) MESCHIA, JAMES F (5.04) GERSZTEN, ROBERT E (5.04) HOWARD, GEORGE (4.61) BRODERICK, JOSEPH PAUL (4.19) PSATY, BRUCE M (4.19) PERREIRA, KRISTA M (3.77) KOTTON, DARRELL N. (3.35) SACCO, RALPH L. (3.35)
1	SCAMPAVIA, LOUIS DANIEL (8.80) NDHLOVU, LISHOMWA C (7.27) HORVATH, TAMAS L (5.24) SPRINGER, SANDRA ANN (4.73) ROSENBERG, PAUL B (4.73) CARRICO, ADAM WAYNE (4.73) BUCH, SHILPA J. (3.71) IWASAKI, AKIKO (3.71) MIMIAGA, MATTHEW JAMES (3.21) SALTZMAN, W. MARK (3.21)
2	MADABHUSHI, ANANT (7.68) OGEDEGBE, OLUGBENGA G. (5.30) MASON, CHRISTOPHER EDWARD (4.82) NIZET, VICTOR (4.82) WANG, GE (4.82) HOU, LIFANG (4.35) EZECHI, OLIVER CHUKWUJEKWU (3.87) NAVAS-ACIEN, ANA (3.87) LARIN, KIRILL V (3.87) ALCAIDE, MARIA LUISA (3.87)
3	GRAY, NATHANAEL SCHIANDER (6.32) WILSON, DAVID M (5.30) LIN, HENING (4.78) WU, HAO (4.27) FELDMAN, EVA LUCILLE (4.27) WANG, YI (4.27) SUH, YOUSIN (4.27) CHENG, FEIXIONG (3.76) YU, HAIYUAN (3.76) KARCZMAR, GREGORY S. (3.24)

Cluster	Top Ten PIs (Normalized Degrees)
4	XU, XIANGMIN (6.88) YEO, EUGENE WEI-MING (5.97) REN, BING (5.51) MILLER, LEE (5.51) GOLSHANI, PEYMAN (3.68) CALHOUN, VINCE D (3.68) ZHANG, KUN (3.68) CHEN, HONG (3.68) WANG, LEI (3.23) XING, YI (3.23)
5	ZHANG, RONG (5.86) BINDER, ELLEN F (5.86) IRIMIA, DANIEL (4.41) STEINBACH, WILLIAM J (4.41) BATRA, SURINDER K. (4.41) MARIUZZA, ROY A (3.92) BURNS, JEFFREY MURRAY (3.44) SEGRE, DANIEL (3.44) DEWHIRST, FLOYD E (3.44) LI, WEI (3.44)
6	BOOKHEIMER, SUSAN Y (5.93) BARCH, DEANNA (5.93) VAN ESSEN, DAVID C (5.12) BENOS, PANAGIOTIS V (4.72) BUCKNER, RANDY L (4.32) AMBALAVANAN, NAMASIVAYAM (4.32) SMITH, STEPHEN MARK (3.92) HEWITT, JOHN K. (3.92) DAPRETTO, MIRELLA (3.92) UGURBIL, KAMIL (3.52)
7	CHUNG, WENDY K (7.21) KIZER, JORGE R (6.19) HOROWITZ, CAROL R (4.15) NEEDHAM, DALE MURRAY (3.64) HOUGH, CATHERINE LEE (3.64) KRITCHEVSKY, STEPHEN B. (3.64) NEWMAN, ANNE B. (3.13) WAPNER, RONALD (3.13) IX, JOACHIM H (3.13) WENG, CHUNHUA (3.13)
8	POWERS, ALVIN C (8.78) KIM, SEUNG K (5.02) KOENEN, KARESTAN C (5.02) RESSLER, KERRY J. (4.34) THOMPSON, PAUL M (4.34) NEALE, BENJAMIN MICHAEL (4.00) SUNYAEV, SHAMIL (4.00) MACDONALD, PATRICK (3.66) BRISSOVA, MARCELA (3.66) ATKINSON, MARK A. (3.66)
9	PERICAK-VANCE, MARGARET A. (7.75) HAROUTUNIAN, VAHRAM (5.61) ROUSSOS, PANAGIOTIS (5.61) BLANGERO, JOHN (5.61) SHEPARD, KENNETH L (4.90) ZHANG, BIN (4.54) HAINES, JONATHAN L (4.19) SCHADT, ERIC E (4.19) LOIS, CARLOS (4.19) SESTAN, NENAD (3.47)

Cluster	Top Ten PIs (Normalized Degrees)
10	KENNY, EIMEAR ELIZABETH (6.52) EICHLER, EVAN (5.28) CLARK, ANDREW G (4.87) MCCARTHY, MARK IAN (4.46) WANG, TING (4.46) GESCHWIND, DANIEL H (4.05) MONTINE, THOMAS J (3.64) LIPTON, RICHARD B. (3.23) FREIMER, NELSON B. (3.23) DABELEA, DANA (3.23)
11	DEBAUN, MICHAEL R. (7.19) WEISS, MITCHELL J (4.81) SNYDER, MICHAEL P. (4.33) STEIN, LINCOLN D. (3.86) STAMER, W DANIEL (3.86) TRAYANOVA, NATALIA A. (3.86) SEEWALDT, VICTORIA L. (3.86) WU, JOSEPH C. (3.86) KHADEMHOSEINI, ALI (3.38) BLOBEL, GERD A (3.38)
12	LI, DEBIAO (7.05) ZHANG, JIANYI (4.70) LENBURG, MARC ELLIOTT (4.70) LEIN, PAMELA J (4.70) SPIRA, AVRUM E (4.70) ABERLE, DENISE R. (4.23) ORALKAN, OMER (3.76) PAGE, JR., C DAVID (2.82) SHIM, HYUNSUK (2.82) WONG, DAVID T (2.82)
13	WOODRUFF, PRESCOTT G (5.68) WENZEL, SALLY E (5.68) LEVY, BRUCE D (4.86) HOFFMAN, ERIC ALFRED (4.46) CASTRO, MARIO (4.46) ISRAEL, ELLIOT (4.46) MEYERS, DEBORAH A. (4.46) DENLINGER, LOREN C (4.05) MAUGER, DAVID T. (4.05) COMHAIR, SUZY (4.05)
14	PLEVritis, SYLVIA KATINA (6.06) DE KONING, HARRY J (6.06) MANDELBLATT, JEANNE (5.34) TEMKIN, NANCY R (4.97) KONG, CHUNG YIN (4.61) HAAS, JENNIFER S (3.88) JIANG, XIAOQIAN (3.51) BIAN, JIANG (3.15) REIMAN, ERIC MICHAEL (3.15) DIAZ-ARRASTIA, RAMON (3.15)
15	ELLISMAN, MARK H (5.62) DICKERSON, BRADFORD C. (5.24) LIANGPUNSAKUL, SUTHAT (5.24) SEUNG, HYUNJUNE SEBASTIAN (4.86) FOROUD, TATIANA M. (4.86) EDEN, URI TZVI (4.48) JOHNSON, STERLING C (3.71) KARN, JONATHAN (3.33) CHEN, WEI (3.33) KRAMER, MARK ALAN (3.33)

Cluster	Top Ten PIs (Normalized Degrees)
16	FIGUEIREDO, JANE C. (6.76) LE MARCHAND, LOIC (5.16) HAIMAN, CHRISTOPHER ALAN (4.77) NORTH, KARI E. (4.37) CONWELL, DARWIN LEWIS (3.97) ULRICH, CORNELIA M (3.97) BELLIN, MELENA D. (3.97) LI, CHRISTOPHER I (3.57) STEEN, HANNO (3.57) HART, PHILIP A. (3.57)
17	LEWIS, JASON S. (6.40) SHAW, DANIEL S (5.35) TERRY, MARY BETH (4.83) GENNARO, MARIA LAURA (4.30) FAYAD, ZAH I A. (3.78) GUO, PEIXUAN (3.78) CANINO, GLORISA J (3.25) MONK, CATHERINE E (3.25) BRITTON, ROBERT A (3.25) ROCKNE, RUSSELL CHRISTIAN (3.25)
18	DIAMOND, MICHAEL S (8.77) LI, JUN (4.21) FREMONT, DAVED H. (3.20) VAN VOORHIS, WESLEY C (3.20) STAPPENBECK, THADDEUS S (3.20) GARCIA-SASTRE, ADOLFO (2.69) DANTAS, GAUTAM (2.69) XAVIER, RAMNIK J (2.69) PADMANABHAN, VASANTHA (2.69) WANG, YONG (2.19)

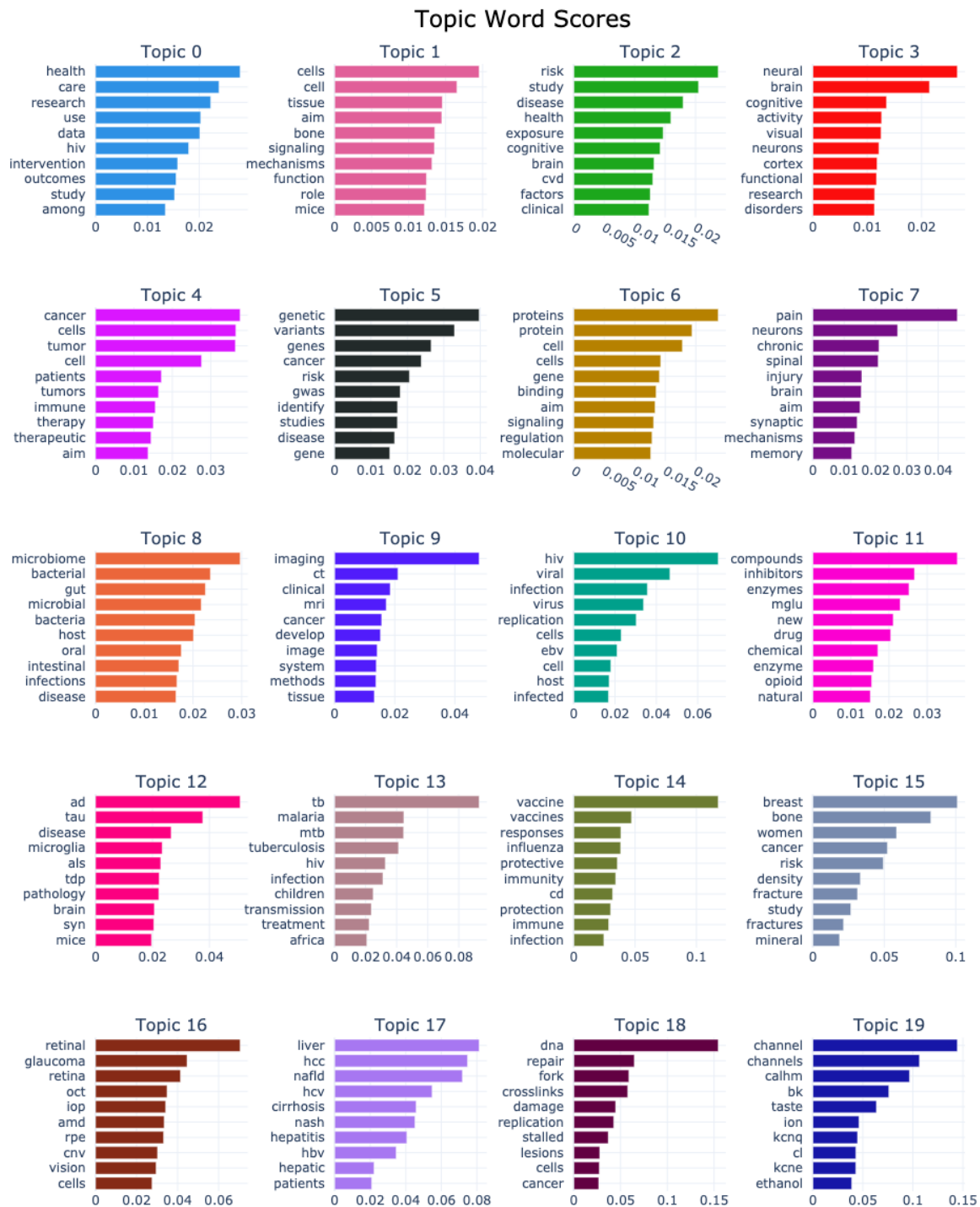


Figure S1: Topic modeling

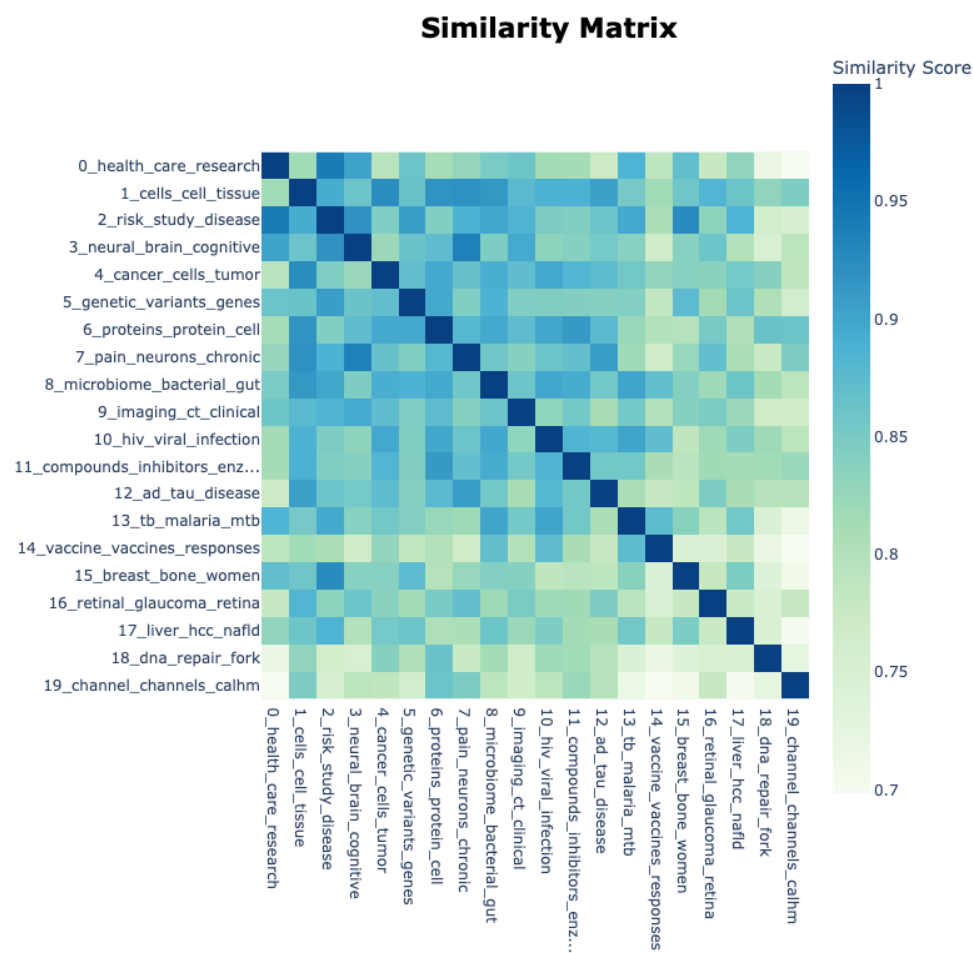


Figure S2: Similarity Matrix

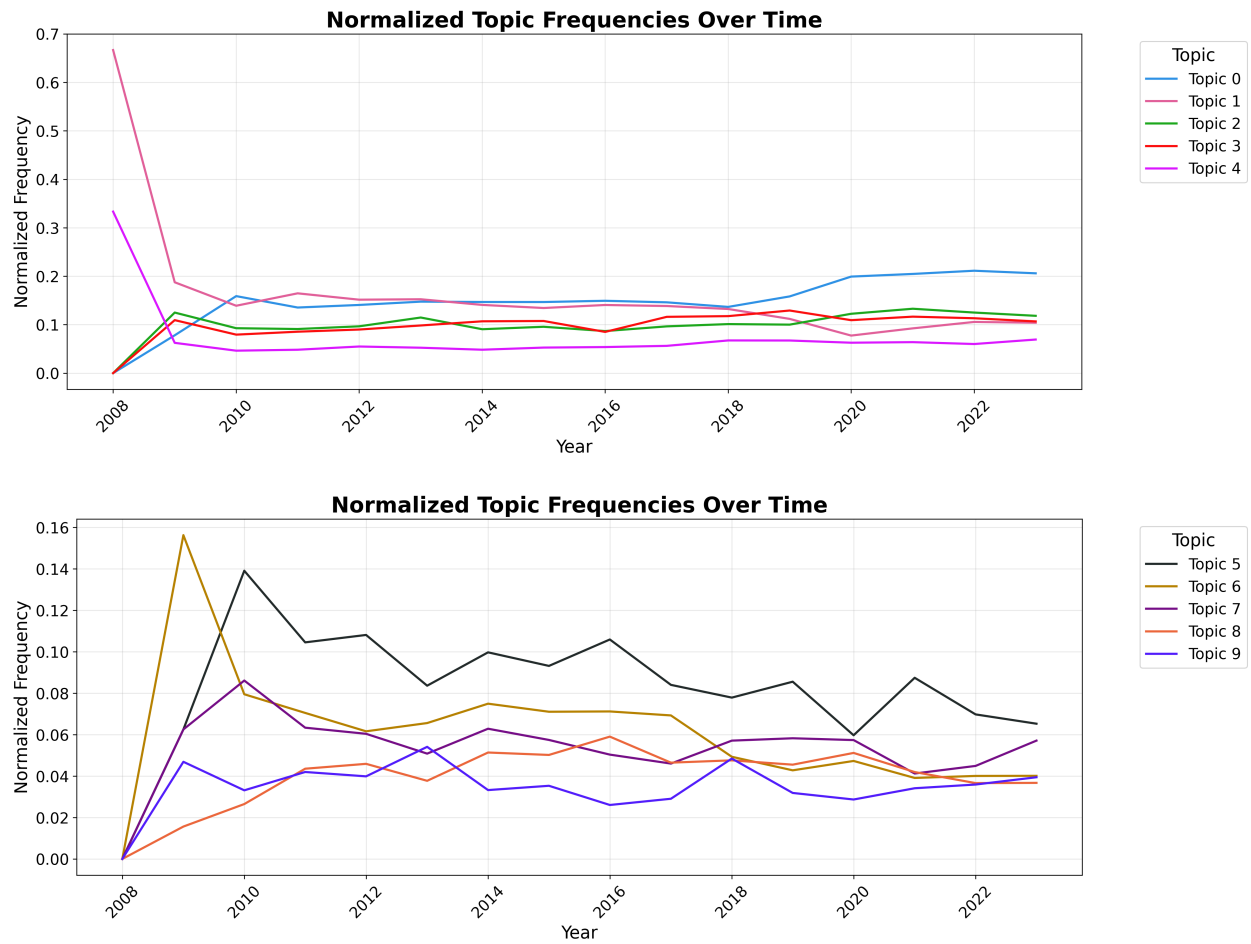


Figure S3: Dynamic Topic Modeling: Topics 0–9

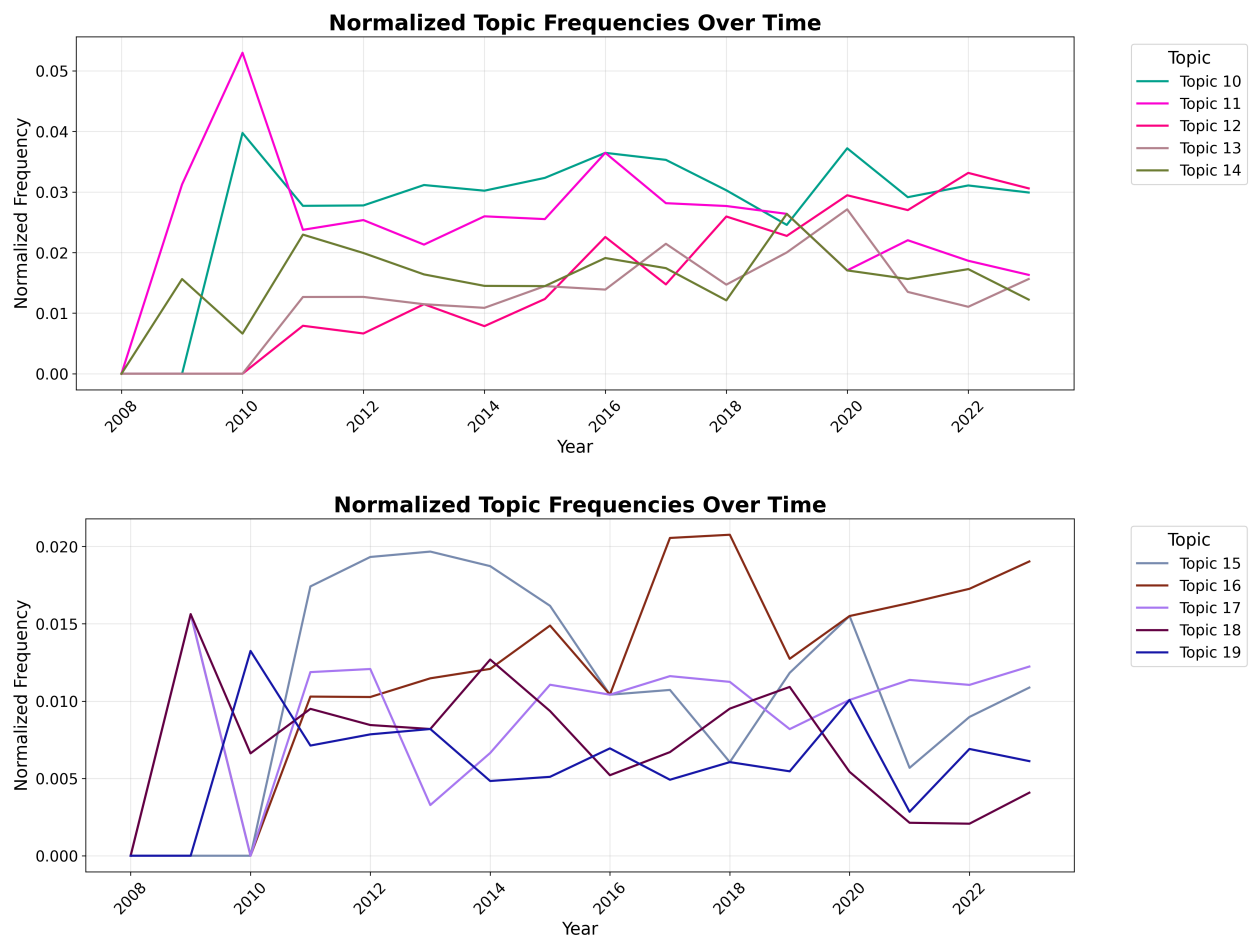


Figure S4: Dynamic Topic Modeling: Topics 10–19

Table S4: Summary of Cluster Characteristics Based on Topic Distributions

Cluster	Top Topics (%)	Summary
C0	T2 (33.1%), T0 (22.0%), T5 (15.2%)	Cardiovascular Health, Outcomes Research, Genomics
C1	T0 (30.9%), T11 (12.9%), T10 (11.4%)	Outcomes Research, Drug Development, Virology
C2	T2 (20.7%), T0 (16.9%), T4 (8.0%)	Cardiovascular Health, Outcomes Research, Cancer
C3	T4 (20.2%), T1 (13.2%), T6 (11.8%)	Cancer, Cell Biology, Molecular Biology
C4	T3 (20.6%), T1 (11.1%), T0 (8.5%)	Neuroscience, Cell Biology, Outcomes Research
C5	T5 (13.3%), T4 (12.2%), T1 (11.1%)	Genomics, Cancer, Cell Biology
C6	T3 (20.5%), T0 (18.2%), T9 (12.7%)	Brain Disorders, Outcomes Research, Imaging
C7	T2 (26.8%), T0 (24.3%), T5 (10.3%)	Cardiovascular Health, Outcomes Research, Genomics
C8	T0 (29.8%), T1 (16.3%), T5 (12.3%)	Outcomes Research, Cell Biology, Genomics
C9	T5 (18.2%), T1 (16.9%), T3 (15.7%)	Genomics, Cell Biology, Neuroscience
C10	T5 (28.2%), T0 (12.1%), T11 (10.0%)	Genomics, Outcomes Research, Pharmacology
C11	T1 (31.5%), T4 (12.6%), T6 (11.9%)	Cell Biology, Cancer, Molecular Biology
C12	T1 (20.0%), T2 (15.2%), T5 (14.6%)	Cell Biology, Cardiovascular Health, Genomics
C13	T13 (14.8%), T1 (12.9%), T4 (11.7%)	Infectious Disease, Cell Biology, Cancer
C14	T0 (40.0%), T2 (18.7%), T3 (13.7%)	Outcomes Research, Cardiovascular Health, Neuroscience
C15	T3 (45.5%), T7 (10.5%), T2 (10.0%)	Neuroscience, Pain, Cardiovascular Health
C16	T5 (31.9%), T0 (17.2%), T2 (10.0%)	Genomics, Outcomes Research, Cardiovascular Health
C17	T0 (22.7%), T3 (18.3%), T4 (10.3%)	Outcomes Research, Neuroscience, Cancer
C18	T8 (17.9%), T10 (16.0%), T2 (11.8%)	Microbiome, Virology, Cardiovascular Health